# Venkata Lakshmi Parimala Pasupuleti

Staff Agentic AI Engineer · Python · LangGraph · LangChain · RAG · LLM Evaluation · AI Safety · Vector Databases · Semantic Caches · Financial Crime · Production AI

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## PROFESSIONAL SUMMARY

Agentic AI engineer with 8 years of production systems and investigative compliance automation. Built **AgentProbe**: 3-agent LangGraph runtime (Orchestrator, RAG Tracer, Auto Fix) with tool calling, task orchestration, semantic caches, and multi-provider LLM routing — plus a 5-metric eval engine with CI/CD regression gates and a React observability dashboard. Built **Leasel**: agentic RAG pipeline (93.6% precision, 1,240 legal PDFs) with a **deterministic rules engine for auditable agent outputs in high-sensitivity environments**. At Entrata: 8 years of **income fraud detection and investigative compliance** (LIHTC/HCV/HUD fake paystub patterns, HAP allocation, AMI threshold automation across 100K+ units) + production AI pipeline on 1.6M financial records. M.S. Data Analytics Engineering, GMU (GPA 3.87, May 2026).

## WORK EXPERIENCE

### George Mason University
Aug 2024 – Aug 2025

**Research Associate — ML Pipelines · Anomaly Detection · Data Modeling**

- Built multi-source data pipelines (Python, R, 190+ countries) achieving **$R^2 = 0.977$**; applied Isolation Forest anomaly detection surfacing anomalies manual review missed across 3 research pipelines; designed analytics dashboards driving **40% improvement** in decision comprehension for faculty leadership.

### Entrata — PropTech / Real Estate SaaS (Opendoor Competitor)
Sep 2018 – Aug 2024

**Senior Software Engineer — Production AI · Income Fraud Detection · Investigative Compliance · AWS · Microservices**

- Built **income fraud detection and investigative compliance workflows** across 100K+ unit portfolios — LIHTC, HCV, HUD fake paystub pattern detection, HAP allocation, AMI threshold automation, rent schedules, tax credit calculations, HUD payment processing, and housing subsidy compliance; zero compliance violations across 6+ years of production.
- Shipped Entrata's **first production AI pipeline on 1.6M financial records** (Python, PyTorch, OpenAI API) replacing manual Excel operations with automated deep-learning workflows; **optimized 200+ SQL queries** (30% performance gain) via execution plan tuning, indexing, caching, triggers, log tables, and audit tables; **achieved 100% PHI compliance** on AWS S3 migration with Libsodium-encrypted pipelines — encrypted 100% of resident data (SSNs, phone numbers), zero security incidents.
- Led **microservices architecture redesign with Docker** — 50% improvement in system scalability, 30% reduction in deployment time; built internal debugging tools reducing debugging time 40%; maintained bug count <10 for 6+ consecutive months; eliminated 80% of manual workload via Python/Bash automation; **300+ code reviews, 5+ engineers mentored**; 100K+ MAU, 99.5% uptime across 120+ Agile sprints.

### Entrata India
Jan 2017 – Sep 2018

**Associate / Trainee Software Engineer — Full-Stack · Compliance Data Pipelines · AWS S3**

- Built full-stack modules for property management and affordable housing compliance (PHP, PostgreSQL, JavaScript); migrated thousands of client documents to AWS S3 resolving 90% of legacy space constraints; processed 1.6M HUD/Tax Credit income limit records via SQL import scripts; created 56 state-specific housing certification documents with AWS S3 integration.

## PROJECTS

### AgentProbe: Autonomous Multi-Agent Framework with Eval & Observability
AGENTIC SYSTEMS · LLM EVALUATION

Python · LangGraph · LangChain · Claude API · OpenAI · Groq · Ollama · FastAPI · ChromaDB · Semantic Caches · Docker · GitHub Actions · React — github.com/venkatalakshmiparimala/AgentProbe

- **Agentic framework with tool use, task orchestration, memory, and semantic caches**: 3-agent LangGraph runtime with tool calling, inter-agent state handoffs, conditional routing, multi-provider LLM routing (Claude, OpenAI, Groq, Ollama); **5-metric LLM-as-judge eval engine** (factual accuracy, hallucination, retrieval, completeness, compliance) with CI/CD regression gates on every commit; React **observability dashboard** tracking per-run metrics, latency, and retrieval drift automatically.

### Leasel: Agentic RAG with Deterministic Rules Engine for Auditable Agent Outputs
AI SAFETY · HIGH-SENSITIVITY

Python · LangGraph · ChromaDB · Semantic Caches · Deterministic Rules Engine · LLM-as-Judge · Pass@k Eval — github.com/venkatalakshmiparimala/Lease_complaince_rag

- **93.6% precision, 87.9% F1** on 1,240 residential lease PDFs — agentic RAG pipeline with semantic caches, context retrieval, and LLM reasoning over legal documents; **deterministic rules engine for fully auditable agent outputs in high-sensitivity compliance environments**; pass@k eval harness (pass@1 81.7%, pass@3 88.3%) with CI/CD regression gate.

### Magizhs: LangGraph Task Orchestration & Compliance Agent
TASK ORCHESTRATION

Python · LangGraph · RAG · ChromaDB · FAA Compliance · Stateful Graph Planning

- Built agentic task orchestration system (parse → plan → retrieve → validate → conditional routing) translating natural-language requests into compliant plans; RAG over FAA Part 107 corpus grounds every decision in cited source regulations; deterministic validation blocks non-compliant outputs before export.

## EDUCATION

**M.S. Data Analytics Engineering** — George Mason University
May 2026 · GPA 3.87

ML systems, AI pipelines, statistical modeling, data engineering, anomaly detection, data governance

**B.Tech Computer Science & Engineering** — JNTU Anantapur
2013 – 2017

## TECHNICAL SKILLS

**Languages & Systems:** Python (expert), PHP, SQL (expert — 200+ production optimizations, execution plan tuning, indexing, triggers, audit tables), PL/SQL, JavaScript, TypeScript, R; System Design, Backend Engineering, Microservices, PostgreSQL, MySQL, AWS (S3, EC2, RDS, Glue), Docker, Libsodium, CI/CD (GitHub Actions), Linux

**AI Agents & LLMOps:** LangGraph, LangChain, Agentic Systems, Agentic Pipelines, Task Orchestration, Multi-Agent Orchestration, Tool Calling, Semantic Caches, Vector Databases, ChromaDB, pgvector (familiar), RAG Pipelines, Context Retrieval, Prompt Engineering, LLM Evaluation, LLM-as-Judge, AI Safety, Observability, Agent Tracing, Claude API, OpenAI API, Groq, Ollama, Knowledge Graphs (familiar), FastAPI

**ML & Data:** PyTorch, scikit-learn, XGBoost, Isolation Forest, Anomaly Detection, Feature Engineering, NLP, BERT, ETL Pipelines, Data Modeling, Tableau, Agile/Scrum, Planning Poker, Git